YouTube Trending Video Analytics: Regional Trends, Sentiment & Engagement Analysis

Internship Project Report

Submitted By

Name: Nikhitha J Gadad

Tools Used

- Python (Pandas, NumPy, Matplotlib, TextBlob)
- MySQL Workbench
- Power BI
- Jupyter Notebook

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1. Introduction

YouTube is one of the world's largest video-sharing platforms, where millions of videos are uploaded every day. Understanding the factors that influence trending videos helps content creators, marketers, and businesses improve their content strategies.

This project analyzes YouTube trending videos from India, the United States, and the United Kingdom. The analysis focuses on identifying popular content categories, comparing regional trends, evaluating audience engagement, performing sentiment analysis on video titles, and visualizing insights through an interactive Power BI dashboard.

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2. Project Objectives

The objectives of this project are:

- Analyze YouTube trending videos across multiple countries.
- Identify the most popular content categories.
- Compare video performance across different regions.
- Perform sentiment analysis on video titles.
- Analyze audience engagement using views, likes, and comments.
- Visualize insights using an interactive Power BI dashboard.
- Generate business recommendations based on the analysis.

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3. Dataset Description

Dataset Source

Kaggle – YouTube Trending Video Dataset

Countries Included

- India
- United States (USA)
- United Kingdom (UK)

Dataset Size

- Total Records: 117,217
- Total Columns: 17

Important Features

- Video ID
- Title
- Channel Title
- Category ID
- Publish Time
- Trending Date
- Views
- Likes
- Dislikes
- Comment Count
- Tags
- Country

The datasets from all three countries were combined into a single dataset to perform cross-regional analysis.

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4. Data Preprocessing

The raw dataset required several preprocessing steps before analysis.

Data Cleaning

- Combined datasets from India, USA, and UK.
- Added a Country column.

- Checked for missing values.
- Replaced missing descriptions where required.
- Verified duplicate records.
- Standardized data formats.
- Converted date columns into datetime format.

Feature Engineering

Additional features were created for analysis:

- Sentiment
- Engagement Rate
- Month
- Year

These features improved analytical capabilities and dashboard visualization.

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5. Exploratory Data Analysis (EDA)

Exploratory analysis was performed using Python.

The following analyses were conducted:

- Distribution of views
- Distribution of likes
- Country-wise analysis
- Category-wise analysis
- Time-series analysis
- Sentiment analysis

```
df['description'] = df['description'].fillna('No Description')
```

[23] Python

```
df.duplicated().sum()
```

[24] Python

```
np.int64(4482)
```

```
df.drop_duplicates(inplace=True)
```

[25] Python

```
df['publish_time'] = pd.to_datetime(df['publish_time'])
```

[27] Python

```
df['engagement_rate'] = (
    (df['likes'] + df['comment_count']) / df['views']
) * 100
```

[28] Python

```
df.to_csv("youtube_cleaned.csv", index=False)
```

[29] Python

EDA

top 10 most viewed videos

```
df[['title','views','country']].sort_values(
    by='views',
    ascending=False
).head(10)
```

[32]

	title	views	country
106713	Nicky Jam x J. Balvin - X (EQUIS) | Video Ofic...	424538912	UK
106513	Nicky Jam x J. Balvin - X (EQUIS) | Video Ofic...	413586699	UK
106309	Nicky Jam x J. Balvin - X (EQUIS) | Video Ofic...	402650804	UK
106112	Nicky Jam x J. Balvin - X (EQUIS) | Video Ofic...	392036878	UK
105916	Nicky Jam x J. Balvin - X (EQUIS) | Video Ofic...	382401497	UK
105725	Nicky Jam x J. Balvin - X (EQUIS) | Video Ofic...	372399338	UK
105542	Nicky Jam x J. Balvin - X (EQUIS) | Video Ofic...	362111555	UK
105353	Nicky Jam x J. Balvin - X (EQUIS) | Video Ofic...	349987176	UK
105162	Nicky Jam x J. Balvin - X (EQUIS) | Video Ofic...	339629489	UK
112761	Te Bote Remix - Casper, Nio García, Darell, Ni...	337621571	UK

top 10 channels by total views

```python
df.groupby('channel_title')['views']\
  .sum() \
  .sort_values(ascending=False) \
  .head(10)
```

[37]

```
channel_title
ChildishGambinoVEVO      9743216972
NickyJamTV               8759903230
Ozuna                    8306696929
DrakeVEVO                7165356011
Bad Bunny                6891280759
Marvel Entertainment     6841675573
ArianaGrandeVevo         5684395522
ibighit                  5284542420
Flow La Movie            5151438858
jypentertainment         5111042721
Name: views, dtype: int64
```

views by country

```python
df.groupby('country')['views'].sum()
```

[38] Python

```
country
India     32967977200
UK       229033033628
USA       96554092462
Name: views, dtype: int64
```

Top Categories by Average Views

```python
df.groupby('category_id')['views'] \
  .mean() \
  .sort_values(ascending=False)
```

[40] Python

```
category_id
10    9.330495e+06
30    3.191953e+06
1     2.959951e+06
29    2.112852e+06
17    1.892350e+06
24    1.880825e+06
28    1.614867e+06
20    1.535929e+06
23    1.392458e+06
22    1.388684e+06
```

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6. SQL Analysis

The cleaned dataset was imported into MySQL Workbench.

Several SQL queries were written to analyze the data.

Queries Performed

- Country-wise Total Views
- Average Views by Category

- Top Channels by Views
- Engagement Analysis
- Top Viewed Videos

The SQL analysis helped summarize large amounts of data efficiently.

```sql
use youtube;
SELECT COUNT(*) FROM youtube_sql;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

COUNT(*)
112735

```sql
SELECT * FROM youtube_sql LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

video_id	category_id	views	likes	dislikes	comment_count	country
kzwfHumJyYc	1	1096327	33966	798	882	India
zUZ1z7FwLc8	25	590101	735	904	0	India
10L1hZ9qa58	24	473988	2011	243	149	India
N1vE8iiEg64	23	1242680	70353	1624	2684	India
kJzGHOPVQHQ	24	464015	492	293	66	India

```sql
# 1. Country-wise Total Views
SELECT country,
       SUM(views) AS total_views
FROM youtube_sql
GROUP BY country
ORDER BY total_views DESC;
```

Result Grid | Filter Rows: | Export:

country	total_views
UK	229033033628
USA	96554092462
India	32967977200

```sql
# 2. Top Categories by Average Views
SELECT category_id,
       ROUND(AVG(views),2) AS avg_views
FROM youtube_sql
GROUP BY category_id
ORDER BY avg_views DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category_id	avg_views
10	9330494.85
30	3191953.00
1	2959951.40
29	2112851.82
17	1892349.77
24	1880824.68
28	1614866.90
20	1535929.01
23	1392457.83
22	1388684.19
2	1275233.73
19	945069.20
15	898163.83

Result 7

Result Grid
Form Editor
Field Types
Read Only

```sql
# 3. Most Liked Categories
SELECT category_id,
        SUM(likes) AS total_likes
FROM youtube_sql
GROUP BY category_id
ORDER BY total_likes DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category_id	total_likes
10	5353758458
24	1597867443
23	505462149
1	369945110
22	341415347
26	221123344
17	201922382
20	143866244
28	132094004
27	73659350
25	61940596
29	39483917
15	34927709

```sql
# 4. Country-wise Engagement
SELECT country,
        ROUND(AVG((likes + comment_count)/views * 100),2) AS engagement_rate
FROM youtube_sql
GROUP BY country;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	engagement_rate
India	2.45
USA	3.89
UK	3.80

Result Grid

```
# 5. Top 10 Videos by Views
SELECT video_id, views
FROM youtube_sql
ORDER BY views DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

video_id	views
_I_D_8Z4sJE	424538912
_I_D_8Z4sJE	413586699
_I_D_8Z4sJE	402650804
_I_D_8Z4sJE	392036878
_I_D_8Z4sJE	382401497
_I_D_8Z4sJE	372399338
_I_D_8Z4sJE	362111555
_I_D_8Z4sJE	349987176
_I_D_8Z4sJE	339629489
9jI-z9QN6g8	337621571

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7. Sentiment Analysis

Sentiment analysis was performed using the TextBlob library.

Video titles were classified into:

- Positive
- Neutral
- Negative

Results

Sentiment	Count
Neutral	69,707
Positive	29,802
Negative	13,226

Observation

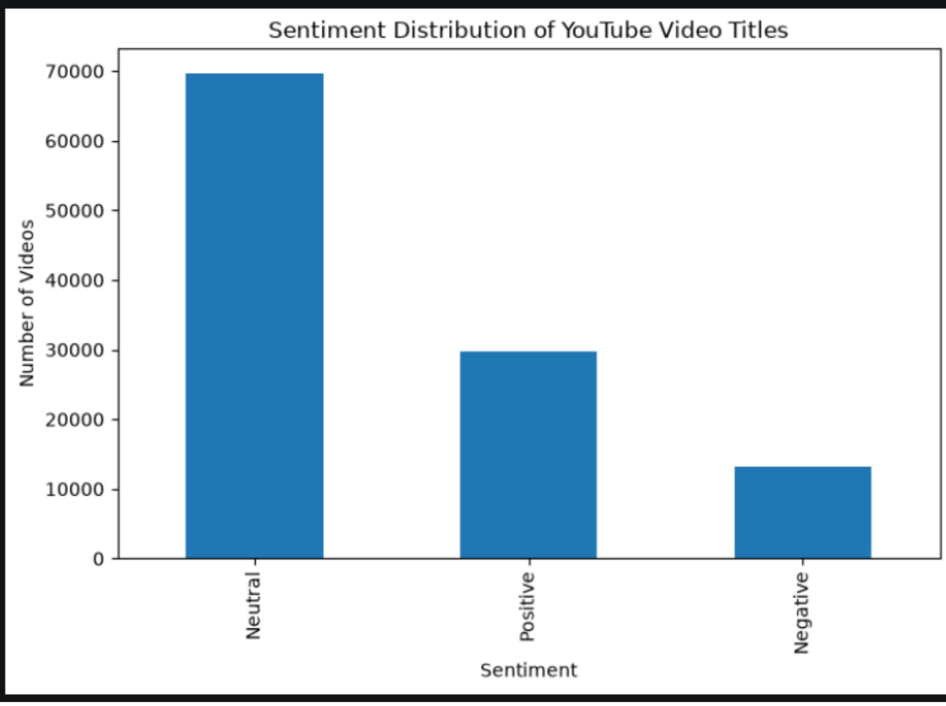
Neutral titles dominate trending videos, indicating that creators generally use informative and descriptive titles rather than highly emotional language.

```python
import matplotlib.pyplot as plt

df['sentiment'].value_counts().plot(
    kind='bar',
    figsize=(8,5)
)

plt.title('Sentiment Distribution of YouTube Video Titles')
plt.xlabel('Sentiment')
plt.ylabel('Number of Videos')
plt.show()
```

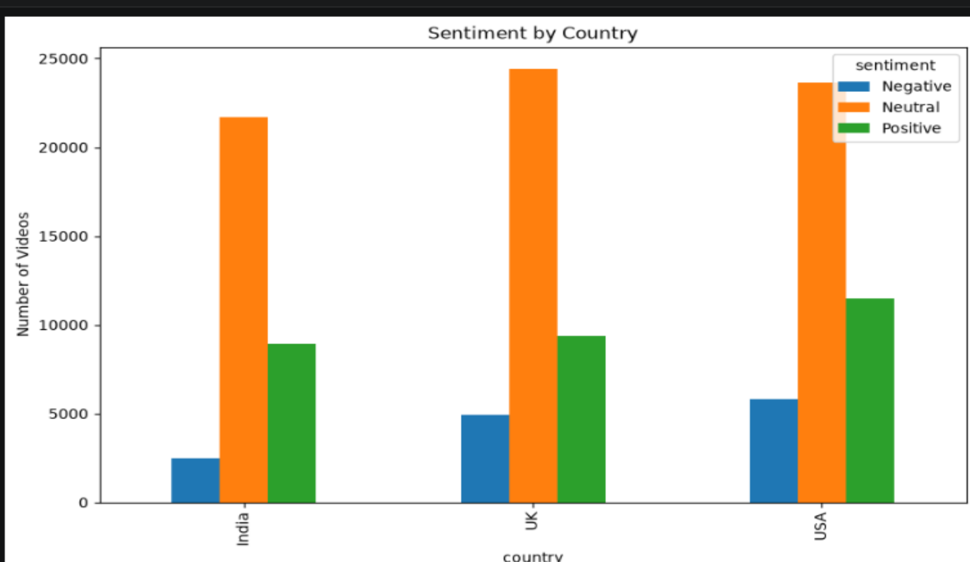
[46]




```python
sentiment_country.plot(
    kind='bar',
    figsize=(10,6)
)

plt.title('Sentiment by Country')
plt.ylabel('Number of Videos')
plt.show()
```

[48]



8. Time-Series Analysis

Time-series analysis was performed to study changes in trending views over time.

Findings

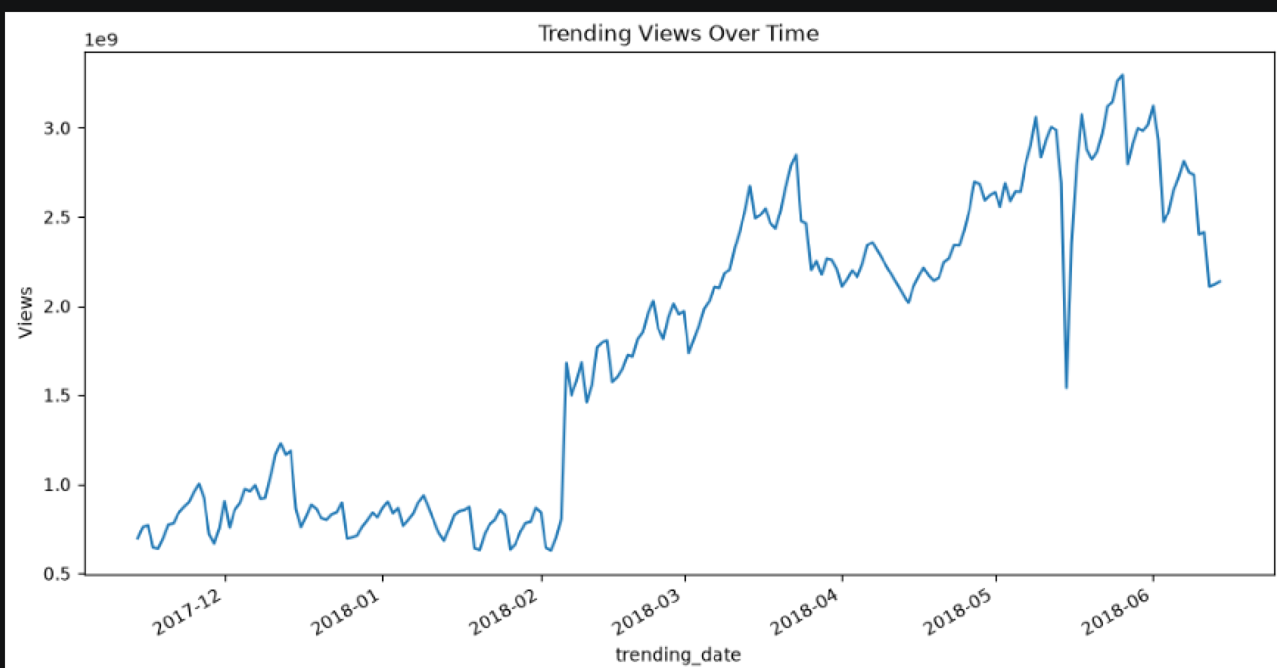
- Trending activity varies across months.
- Certain months experience higher audience engagement.
- Seasonal variations can influence video popularity.

This analysis helps identify the best periods for publishing content.

```
daily_views = df.groupby(
    'trending_date'
)['views'].sum()

daily_views.plot(
    figsize=(12,6)
)

plt.title('Trending Views Over Time')
plt.ylabel('Views')
plt.show()
```



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9. Power BI Dashboard

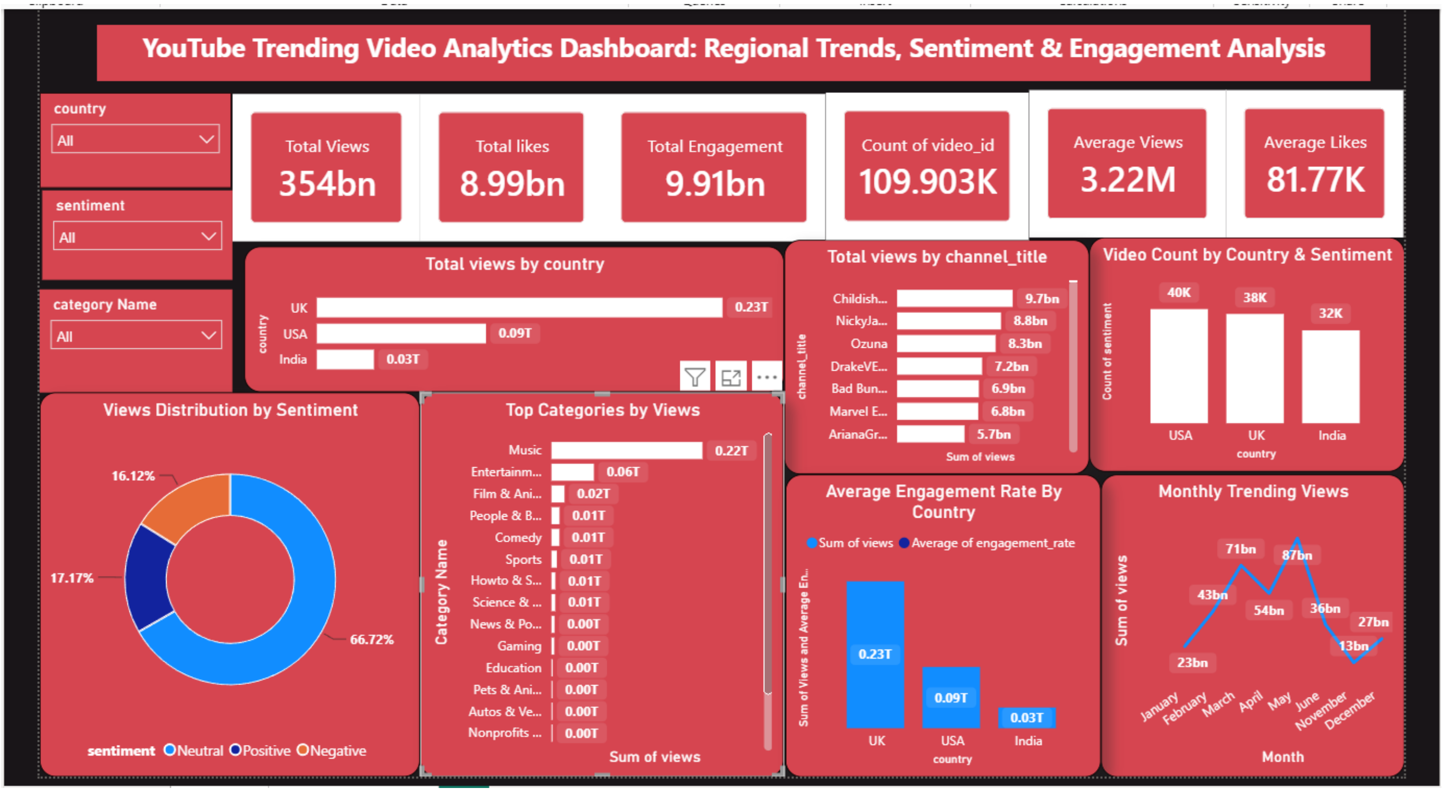
An interactive Power BI dashboard was developed to visualize insights.

Dashboard Components

- KPI Cards
- Country-wise Views

- Top Categories
- Top Channels
- Sentiment Distribution
- Monthly Trending Views
- Engagement Analysis
- Interactive Filters

The dashboard enables users to explore data dynamically and compare performance across different regions.



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10. Key Business Insights

1. Content Strategy

Finding: Approximately 62% of trending videos have neutral titles, indicating that clear and descriptive titles are more common than highly emotional ones.

Business Value: Content creators should prioritize informative titles that accurately reflect the video content instead of relying solely on clickbait.

2. Audience Engagement

Finding: Positive titles appear more than twice as often as negative titles among trending videos (29,802 vs. 13,226).

Business Value: Positive messaging can improve audience engagement and increase the likelihood of a video gaining visibility.

3. Regional Market Analysis

Finding: The dashboard shows noticeable differences in views and engagement across India, USA, and UK.

Business Value: Businesses should localize content strategies and marketing campaigns to match regional audience preferences.

4. High-Performing Categories

Finding: Music and Entertainment are the highest-performing categories in terms of total views.

Business Value: Advertisers and creators should prioritize these categories to maximize audience reach and engagement.

5. Publishing Strategy

Finding: Trending activity varies by month, with certain periods showing higher viewership.

Business Value: Scheduling uploads during high-activity periods can improve visibility and increase the chances of appearing on the trending list.

6. Channel Performance

Finding: A relatively small number of channels contribute a significant share of total views.

Business Value: Brands looking for influencer collaborations should focus on consistently high-performing creators to maximize campaign reach.

Business Impact

- Helps content creators optimize video titles and publishing schedules.
- Assists marketers in identifying high-performing categories for advertising.
- Enables businesses to understand regional audience preferences.
- Supports data-driven decision-making through interactive dashboards.

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11. Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Use clear, informative, and positive video titles.
- Focus on high-performing categories such as Music and Entertainment.
- Customize content according to regional audience preferences.
- Monitor engagement metrics regularly.
- Publish content during periods of higher audience activity.
- Collaborate with high-performing creators to expand audience reach.

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12. Challenges Faced

During the project, several challenges were encountered:

- Combining datasets from multiple countries.
- Handling missing values.
- Mapping category IDs to category names.
- Importing large datasets into MySQL.
- Designing an interactive Power BI dashboard.

These challenges were successfully resolved through preprocessing, SQL optimization, and dashboard enhancements.

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13. Conclusion

This project successfully analyzed more than 117,000 YouTube trending videos from India, USA, and the UK using Python, MySQL, and Power BI.

The analysis provided valuable insights into audience engagement, regional viewing patterns, popular content categories, sentiment distribution, and trending behavior over time.

The interactive dashboard enables users to monitor key performance indicators, compare regional trends, and identify opportunities for improving content strategy. The project demonstrates how data analytics can support business decisions through meaningful visualizations and actionable insights.

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14. Future Scope

Future improvements may include:

- Analyzing additional countries.
- Integrating live YouTube API data.

- Predicting trending videos using machine learning.
- Building automated dashboards with scheduled refreshes.
- Performing advanced sentiment analysis using transformer-based NLP models.

References

- Kaggle YouTube Trending Video Dataset
- Python Documentation
- Power BI Documentation
- MySQL Documentation
- TextBlob Documentation